

Active Inference Stream #009.1: "The Projective Consciousness Model and Phenomenal Selfhood" (2018)

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hello and welcome everyone to the active inference live stream it is act imp stream 9.1 and it is November 24th 2020 so welcome everyone to the ACT imp stream and welcome to teamc today in the ACT imp stream we will be talking about a really interesting paper so I'm quite excited tecom is an experiment in online team communication learning and practice related to active inference you can find us at our website our Twitter at email our YouTube or our public keybase team and shared key based username this is a recorded and an archived live stream so please provide us with feedback so that we can improve on our work all backgrounds and perspectives are welcome here and also just as far as video etiquette for live stream remember to mute if there's any noise in your background raise your hand so that we can hear from everybody on the stack and use respectful speech Behavior Etc just to quickly go through the rest of the meetings that we'll be having for 2020 um today on November 24th we're talking about the projective Consciousness model and phenomenal selfhood next week in 9.2 on December 1st we're going to be talking about the same paper so anything that gets raised today whether you're participating live or watching a replay let us know about it so we can cover it in 9.2 on December 8th and 15th in 10.1 and 10.2 we'll be discussing the paper a variational approach to scripts and then on December 22nd and 29th we'll be talking about sophisticated effective inference and that will be another more on the modeling side paper to discuss cool today in act mstream 9.1 the road map will be as follows we'll start off with some introductions and warm-up questions and then we'll get to 9.1 we're going to be talking about the paper called the projective selfhood by Williford at all 2018 and the punchline of this paper is basically about the combination of a few different fields geometry cybernetics neurophenomenology Etc and by combining these fields under the framework of active inference there might be something there to understand or explain about Consciousness so we'll go through the aims and the claims of the authors the abstract and the road map and then

at multiple points here and throughout we want to take time for questions so really anyone who raises their hand will just as soon as we can um cut there because it's so interesting what people have come to mind during these provocative topics then we'll go to the author's study of Consciousness and talk about the two lists that they make and what those lists do for the paper and do for their Theory and those are the functional features and the phenomenological invariance of Consciousness and then at the end we'll go through the figures and with however much time we have we'll go into thinking about what each figure might represent or invoke and as stated in 9.2 we're going to be discussing this same paper again so save and submit any questions you have and get in touch with us if you want to participate great for the introductions and up please just introduce yourself and your location say hello and as short of an intro as you'd like and then pass it to somebody else who hasn't yet spoken so I will go first I'm Daniel I'm in California and I'll pass it to Alex uh I hope I'm audible my name is Alex I'm also in California uh and I will pass this to uh Shannon hey I'm Shannon I would be in California normally but I'm in South Dakota so I'll pass it to Lee hi I'm Lee I'm in London uh and I'll pass it to Mattis R currently muted okay hi I'm Mathis I'm currently in Germany in AAR Brook um and I passed it to even hello my name is vanan I'm from Moscow Russia and I pass it to Sasha hi my name is Sasha I'm also in California and I'll pass it to Alex uh V thanks hi I'm Alex I'm in Moscow Russia and I pass it to Steven hello I'm Steven I'm in Toronto in Canada cool that is everyone so great thanks for these introductions everybody let's do some warmup questions and there'll be a few of them and anyone who would like to talk just jump on with raising your hand let's try to hear from everyone as well the first question and I'll do actually the first and the second question together since I think that they're very related the first question is what interests you about studying Consciousness and this second question is what are some ways that we can study Consciousness um maybe someone's a skeptic maybe they don't think that we can study Consciousness or maybe they have a really awesome uh idea about how we could do that so Lee uh go ahead what's your first thought uh yeah so um I mean I primarily come from the phenomenological angle and um particularly kind of neurophenomenology really and what I'm really interested in is agency so can the can the models uh that come out of uh coming out of Neuroscience can they help us to um increase our our agency so for example um if you can become reflexively aware of simulations modal simulations that are going on in your system then do those become affordances for actions that we can take and so you increase the scope of of agency so that's kind of what really interests me about this cool thanks Shannon I love that you bring up agency right off the bat um

something that's really interesting for me about Consciousness which I don't study at all is when we lose our sense of agency like when we're in flow States if we're making music or I guess people get in flow States if they really like exercising and we're completely immersed in the experim experience but not conscious aware or we don't have the same kind of intentionality that we do as if maybe we're we're like writing down a math problem or trying to speak over a live stream and and thinking very carefully about each of our words and I get guess just the phenomenal quality of those two types of experience is really interesting to me I agree and it's part of this what is consciousness question and what is the role of awareness so when we're really paying attention to doing a task with with our hands when we're really paying attention to balancing on a beam or something like that are we truly at our most self aware at that moment or are we at our least self-aware or in which ways are we self-aware we're not thinking about our biographical history per se but in another sense we're the most embodied and anactive sort of circuit for Action so where is that going to fit in with active inference and with this idea that we are embodied in an environment so what is yeah really more experiential of the self doing something that might take your whole body or bring your full attention to bear or doing something that asks you to write a personal statement something like an autobiographical or narrative self cool if anyone else wants to go on the first two those are good questions or the last question here is just like what is another theory or framework for Consciousness that you know of or have heard about doesn't need to be your theory to um support or deny here just like what is something else that is in the space of this paper and then what's just something that you liked about it or found absent from that model so all just starts with something that I find missing from a lot of models and I really want to know where agency and where a lot of other topics fit into play here is it's often a little unclear whether Consciousness is something that reaches back into the system and intervenes or whether it's something that's just purely Epi phenomenal or kind of um just there could be awareness but awareness may not have causal power so is consciousness something that's just getting spun off to the side and produced but not reaching back into the system or is it something that's actually essential potentially even in an evolutionary or adaptive context for maintaining coherent Behavior Stephen yeah I'm I'm interested in in this question of self-consciousness that seems to be um and I like just the idea of General Consciousness being thought about because I think in psychology this idea of the self and the idea of the individual has become so dominant and the idea of broader Consciousness was kind of left a 100 years ago for the the philosophers to deal with so this idea of I have a self and I have this um idea of a psychology

and this perspective on the psychology has become kind of all-encompassing but now this ability to bring Consciousness in more broadly sort of dissolves a lot of those problems because it it kind of stuck psychology at this single scale analysis um so that's one of the things I think that um is quite exciting by all of this cool Sasha um one thing that I I find missing I feel like a broken record sometimes um is the developmental side of Consciousness um and even if we're thinking about humans you know our favorite species uh how does Consciousness develop across a lifetime time and what are the different uh stages or versions of Consciousness that we pass through um and is there sort of a developmental trajectory because um I tend to think it's not like flipping a light switch think it's not that we are born and we're immediately plugged into this um same Consciousness that we're experiencing today so um yeah that's kind of what I would like to see from a Consciousness model is different levels or um versions of it that we can start to describe like much like the Flow State sleep cool Shannon yeah I like both of those um points that Stephen and Sasha brought up and not just how does our our personal Consciousness develop over time but how does our Consciousness like our group Consciousness um develop so like as an infant a lot of your conscious experience is really coupled with uh the mother or the caregiver and like slowly you might develop a sense of of yourself as an individual separate from that caregiver but also when We Gather in groups you know when We Gather at concerts or We Gather at protests or in events where we're mix of explicitly or implicitly interacting with each other we start to behave like like one group and in some sense we start to really feel like one group rather than like just myself in a room full of other people yep cool and something in the ant literature people have wondered about is the nestmate conscious is just the colony conscious but not the nestmate then what is really happening with this transition from Individual to group and the development of that also interesting the formation of group is it kind of like how many sand grains make a heap you know once there's three people it's a collective Consciousness or do they have to be doing contact improv or do they have to be rhythmically sing synchronized what are those Dynamics what might be metrics what might be things that we could come to the table and agree on as far as measurements are concerned so that we could kind of move the needle forward in research or is this a question that because of its maybe fundamental difficulty in measurements is it a question that's sort of forever going to be in the speculative domain I think that the authors of this paper would suggest that scientific study of Consciousness is Meritus and that there's value in approaching with even a mathematical bent approaching how we could study Consciousness though it's by no means obvious that

mathematics is the way that we're going to express the language or the structure of Consciousness especially when we start to think about all these different dimensions like group versus individual and what you know who is individual we were never individuals that type of thinking what does the brain region know that the neuron doesn't and the Brain know that the region doesn't all these questions are going to come into play and a lot of the the neuro and social parts are going to get a little bit um not swept under the rug here but this paper is clearly a little bit more on the cut and dry projective geometry side and there's a lot of discussion of neurophenomenology but you don't see any pictures of cells you don't see any actual details of neurom mechanisms it just assume well you could wrap mechanisms and mechanisms kind of close to the end uh to get this type of A System's behavior and so we'll get a little bit more into it so at any point of course people are welcome to raise their hand we'll just try to immediately whatever Springs to mind in a relevant moment so no worries about these prompts let's talk about the paper which is the projective Consciousness model and phenomenal selfhood by willford at all in December 2018 in Frontiers in Psychology theoretical and philosophical psychology the aims and the claims of the paper are basically as such they ask can we describe this integrated multimodal and centered spatial structure of conscious experience in a rigorous way and can we give a good account of why it should be thus organized here we answered both questions in the affirmative great big if true let's see these be answered overall the PCM delivers an account of the phenomenologically available generic structure of Consciousness and that speaks to Steven's Point about the general Consciousness and shows how Consciousness allows organisms to integrate multimodal sensory information memory and emot in order to control Behavior enhance resilience optimize preference satisfaction and minimize predictive error in efficient manner and this is again to my earlier point about does Consciousness get spun off of the system or does it reach back in so does Consciousness control behavior and enhance resilience or is that systems with controlled behavior and enhanced properties like resilience do those systems spiral off Consciousness as an observer that's vague here but the way that we phrase these sentences is is essential it's like active versus passive mistakes were made versus I made a mistake versus you made a mistake pretty different and the non uh jargony way might be to ask how can we apply active inference as this unifying framework to the study of Consciousness in all of its unified coherent mysterious Dynamic and profound facets and we'll see when they get to the phenomenological invariance which aspects of Consciousness they think are sort of the key features to explain uh and really integrate across into one common framework so that's what

we hope to be getting at today even though this paper is not the whole citation Network we're definitely going to be exploring a lot of the key ideas and questions about how and where active inference and the free energy principle could link back to Consciousness so here's the abstract briefly they write we summarize our recently introduced projective Consciousness model PCM which is rud at all 2017 and relate it to outstanding conceptual issues in the theory of Consciousness the projective Consciousness model combines a projective geometrical model of the perspectival that means from a perspective phenomenological that means what's experienced structure of the field of Consciousness with a variational free energy minimization model of active inference comma yielding an account of the cybernetic that means goal-seeking function of Consciousness viz the modulation of The Field's cognitive thought processes and affective like how it feels emotions Dynamics for the effective control of embodied agents another uh plus one for affective and effective in the same sentence abstract part two the geometrical and active inference components are linked via the concept of projective transformation so we'll definitely have a slide where we look at what a projective transformation is which is crucial to understanding how conscious organisms integrate perception emotion memory reasoning and perspectival Imagination in order to control Behavior enhance resilience and optimize preference satisfaction the PCM makes substantiative empirical predictions and fits well into a neuroc computationalist framework it also helps us to account for aspects of subjective character that are sometimes ignored or conflated pre-reflective self-consciousness the first-person point of view the sense of mindness or of ownership and of social self-conscious so maybe we will get to some of these social topics and they close by saying that we argue that the PCM though still in development offers the most complete Theory to date of what Thomas metzinger has called phenomenal selfhood so that's combining this phenomenology component with a selfhood uh element any thoughts on the abstract or we can move forward bring them up during the road map so the abstract is just the header of the paper it's kind of like the metadata for the ingroup and outgroup discussion the road map is how they're actually going to implement these kinds of claims that are suggested in the introduction in the abstract so first with a more formal introduction to the topics they discussed in the abstract and then they went straight to methodology which I think is very important especially when discussing something that's abstract like this then they talk about the phenomenological invariance and the functional features of Consciousness which we'll go through on separate slides and specifically they talk about these invariance in terms of a projective geometry feature of the PCM they talk about the functions and then talk about

how the PCM links the function features of Consciousness within an active inference and a cybernetics framework so they're taking some of the invariants combining them with projective geometry they're taking some of the features and combining that with cybernetics uh in the sense of active inference as a modern phrasing of cybernetics and free energy minimization as an optimization process and they're kind of connecting experience cybernetics projective geometry all these different things together and asking whether it makes sense they then have all their four figures which we'll go through today that range from sort of sketch-like accounts of psychological phenomena to some early solutions to otherwise challenging ideas like uh Illusions necter Cube and then they sketch out a few different representations and simulations that they run or ran elsewhere they then return to this question of the hard problem representationalism and phenomenal selfhood so the hard problem is the hard problem of Consciousness attributed to Chalmers and others that relat to basically where the rubber hits the road how does Consciousness arise or what is it I think there's many hard problems uh and who knows which ones we think are easy but actually are hard the point is they're going to come at that Nexus from a computational and a functional angle then they talk about representationalism in the PCM and I feel like representationalism we've heard about a bunch on active stream with a basian structural representationalism and now we're going to come at it from a different angle from a geometric and a cybernet angle they then talk about how the PCM is linked to subjective character as well as phenomenal selfhood which is clearly one of the things that they're out to describe and then they conclude so they're going to go from taking a few different threads of phenomenological neuro research and tie that with some more mechanism agnostic control theory approaches like cybernetics active inference and optimization approaches like free energy minimization they're going to bring in this projective conscious uh projective geometry bit to tie it all together and then we get to the conclusions and also some of their thoughts on where to go next so let's just pause there I've written out a few questions but does anyone just have a thought or a question or when they read the paper like what was a first thought was there something that resonated a lot was there something that seemed IL defined or you would have defined it a different way or just something you curious about Stephen well one thing I thought was interesting was this idea of perspectival Consciousness so this idea that when you're consciously taking a perspective on your situation um which you normally might think of as seeing it but it could be some sort of embodied sense so I think that was kind of important and there and how that differs from maybe the unconscious um dynamics that are just going on that they don't necessarily deal with so it's

somehow sitting I don't know if it's sitting on top or sitting alongside it but there's there's somehow this perspectival piece that somehow kicks in yeah and we often think about Vision in talking about how active inference is to be differentiated from let's just say a sensory relay model where you know photons come in your brain categorizes them and makes an image and then you act and here it's like it's a generative model of the world that you're experiencing that's why there's color and good resolution around the edges and no blind spot and then it's like well then why do we see what appears to be coming from our eyes or from a position that's right there and that's where we're going to come back to with a lot of these questions about the projection itself and why we appear to be situated Lee sorry yeah hi I don't know if it's relevant but um I was just thinking about so the the discovery of perspective uh you know back in the in the Enlightenment and I was thinking well obviously you know if this is a sort of a more generic Theory Of Consciousness then um I was wondering if it connected perhaps to met singer's idea of transparent States so that uh you know the projective geometry would have always been there but they would have been transparent and somehow you know in the process of the discovery of perspective uh it became something that we were able to pay attention to more more consciously anyway just a thought cool I IDE let me actually unpack that first what would you describe as the transparent State because I don't know of that terminology specifically what is a transparent State here so I think in met singer's model so transparent States would be those states that we are that we view the world through so we that we're in a state but we don't know that we're having a particular State whereas he he contrasted to a metaphor I think really so an know P state would be a state where we where we understand that it's a a um a sensory state if you know what I mean cool the transparent State it's kind of like what is water the water we swim in it's like the water is transparent so when you look out all directions it's not being seen specifically um yeah or glass like you're looking through it you're looking through it but you're not able to actually observe it as a state of of your as as a state an embodied state that you're having if you see what I mean yeah just to unpack this one bit for Stephen um it's kind of like without a theory of air you could just look forever and you could just have any model could replace this idea about what fills up space then there's this chemical theory of air and there have been various chemical theories of air and whether there's four types of things and they mix or whether it's protons and neutrons and different ratios and things like that and then once you have a theory of air you could ask things like well what if I compressed all of the quote invisible stuff in this room and all of a sudden it looks like blue or something it's like whoa it kind of is there there's unique experiments there's

unique predictions I'm taking something that was transparent in the nich that actually was real ontologically real and then through attention and awareness and affordances we're able to now like look at it so we've developed projective geometry and it's like well what if it really were projective geometry all along but then now with the projective geometric tools we're coming back to experience that and it's actually something that Stephen and I were talking about just this weekend with um a little bit like what people's experiences of perspective have been in the past um because there is those sometimes funny pictures um at least comical sometimes where it's like a person's face and it looks like it's from a almost a Cubist angle but it's like a Renaissance painting and there's probably so much to say from an art history perspective about the evolution and the development of perspective but at the very least it seems like people throughout time have been differentially excited by different types of representations of perspective people weren't doing photorealistic charcoal drawings in a different era even if they necess could have had the um Technologies to do so so something's happening there with perspective and how it's changing the way that we interact with a visual World cubism of course was shocking and uh when it came out and so now where are we with perspective um there's a lot to say about that Stephen yeah I mean one of the things that's interesting with like imagined feelings in space is that they can be kind of transparent so if you like what's the energy of someone or if I imagine the energies of people around me it's like a quality those qualities they don't like if you imagine something in space it doesn't become completely opaque it's not like you can't see that you're in the room anymore you know so even if you're on stage as an actor and you imagine that you're in a scene with a king and it's an empty stage and you start to imagine the whole stage you it's still slightly transparent in relation to the physical perceptive conscious physical space if that makes sense so those two things are kind of interplaying yep yeah and and to bring that to the affordances the field of affordances field of Consciousness it's like the generative model isn't just of what the video camera would see if it were next to your head the generative model is actually generating this field of action that's scaled geometrically so that things that are closer to you and better within reach are seemingly at hand and things that are distant are seemingly distant and one example that Stephen and I were talking about was like there might be an object on the other side of my monitor that I'd have to kind of go around to reach and just right with a quick judgment it feels like something on the other side of my monitor is actually further away than something that might be linearly closer um or further away sorry but like I can reach it directly just by reaching my hand over I can reach two feet over with my arm just grab it but the 18 inches linearly to the monitor I

have to end up doing a 36 inch route so it feels longer away so even within our atand Concepts and the way that we talk about what is accessible or what is um relevant these all have to do with the grasping and the comprehending and the action Distortion of a space so there's really a lot here about the perspectives that we take and if we say there's one perspective and it's mine there's one perspective and it's my cultures or is it hey we're all at this party and we're all going to have different perspectives of this party okay which perspectives can hang which ones can't those are the kinds of discussions that you get to there instead of is it right wrong the way I'm seeing it you know not everyone's going to see the party like you see it because no one's head is overlapping with your perception because everyone's affordances and location are different for example Sasha yeah I really appreciated in this paper um the kind of connection um uh we're always talking about on this stream um between p uh you know taking in information and acting um and so it's this feedback loop of two things and I really like this converting uh thinking about our space into thinking about the space of affordances so that it's not just uh a you know a blank slate of information that uh you could access or not but it's really what you are able to do with that information and how you're going to act on it um so I yeah I just thought that was a really nice connection and fits in very well with um what we're talking about and kind of builds up this um similar theme of how you're going to connect action and perception cool and just one other example that that kind of uh made me think about it's like when I'm dissecting an insect under a microscope um it's very small movements of the fingers but it in the field of vision of a microscope will swing the forceps like you know 30° so it'll be a radical motion under the microscope but it has the exact same embodiment feel as like swinging one's arm almost as if what matters is how much I'm seeing my body move within a visual field and how radical of an affordance it is like flailing my arm versus a more fine-tuned movement but a fine two movement with the arm is like one inch but that's represented by a certain number of degrees and that ends up being like visually almost the same number of degrees that I can move the forceps so even though it's at a different spatial scale because the attention and the affordances are down at that scale and I have the visual um ability the umelt to perceive at that smaller scale which couldn't be seen if it were the nak eye it takes on a feeling like it's in the regular Vision regular body world and then also the uh forceps don't have nerves but they're sensitized so there's a lot to say and think about there um a few questions that I just wrote down here just like these are of course perennial questions they're not going to be answered in a couple minutes but what's exciting about studying Consciousness what do we agree on what do people agree on what do people

disagree on and then also what do people agree is unknown what are the known unknowns where might there be unknown unknowns and then also to always return to what is active inference doing here what could it do here what could some other Theory do here what do you think Shannon yeah so your story about the forceps in the microscope reminded me of the first example of like extended Consciousness that I learned about in in a philosophy of Mind class in undergrad and our professor was talking about divers um who are they're not actually diving they're in a room and they're controlling a camera like on a on a Little Submarine that's underwater and they start to feel like they are the the submarine so much so that their body is actually responding to the the pressure differences that they would experience if they were diving so if the submarine comes up very fast they'll actually feel like they're experiencing the bends that would happen if they were a diver coming up so fast I don't know if this is a true um story I've never looked it up but it's very intriguing that your Consciousness can be can be so extended that you end up having these embodied effects that you perhaps could empirically measure if you if you were if you had some sensors on on these people as they were experiencing these extended conscious moments cool Lee um I was just going to ask Shannon um know whether that depends on whether the people controlling the remote subware have previously dived as well so they kind of got the you know the embodied experience as part of their model I think for that for specifically the bz it would or maybe they would have to know about the B because that's a very specific thing that doesn't happen to your body in any other case yeah one thing that makes me think of is like uh the drop in a techno song when everything's building and the tension's building but that's actually a cultural Reg regime of affordance and different subg genres of Electronica or different people who are aren't familiar with Electronica they might get chills going down their spine it's like oh this is the sickest drop ever but somebody else might just not be tuned into that regime of attention and therefore it's like okay wait what happened why is it just why did the drums stop something like that so similarly it's like the diver if they're part of this deep generative model and then they're they're emulating it or they're simulating it somehow um this diving then they may have a deeply learned reflex about changing their blood pressure let's just you know go with st experiment being vaguely you know feeling true just is that you could if you're doing a first-person shooter immersive video game your heart is going to start pounding not because you need increased blood flow that's a psychological response so trained psychological responses in increasingly immersive environments like AR and VR will generate physiological responses and that is actually part of the thrill that was I mean there's so much happening here and it

doesn't really go into the AR VR here but like what if we unprojected our projective geometry what would happen or what about these uh sort of Allison Wonderland type simulators for the AR and the VR where you reach towards something but your arm moves twice as fast or you reach towards something and your arm doesn't move and you crash into stuff and it just feels like you're like super disoriented or the glasses are flipped all of these Transformations are just super deep interventions into the structure of sense and action so there's a lot that is now possible here with how we can on fly remap these things in 1920 the Cubist could put a light on one side of the cube and draw it and then halfway through the drawing move the light and to get a different perspective and that was kind of cool like a mashup of perspectives in one two-dimensional plane but now you could be experiencing like a lot more stuff happening and the cube could be mutating it could be hyper Cube and stuff like that it's kind of next level with respect to our geometry and how it relates to our sense and action any other thoughts or questions here before we go to the functional features just think of them as they come up so the authors highlight four functional features of Consciousness which are Global optimization and resilience Global availability motivation of action and modulation of attention and simulation enhancement so their goals are basically these are the things that they say it's notice that it's very mechanism independent it doesn't say you got to have neurons you need Gia it doesn't say it has to be at this time scale these are all like things that you might say about a cybernetic system a control theory action perception system so these features are the abstract functional features of Consciousness which through neurophenomenology will get brought into the brain perhaps but just at the functional level these are the functions of Consciousness and they write we claim that the overall function of Consciousness again is it that Consciousness is getting spun off by systems that have these functions or is consciousness literally the function that Maps back into the system and gives these features um presence in the world we claim that the overall function of Consciousness is to address a general cybernetic problem or problem of control Consciousness enables again ambiguous a situated organism with multiple sensory channels to navigate its environment and satisfy its biological and derived needs efficiently this entails minimizing and predictive this entails minimizing predictively erroneous representations of the world and maximizing preference satisfaction a good model of Consciousness must be able to explain how this is accomplished the PCM thus emphasizes the following interrelated functional features of Consciousness so ask yourself if there was a robot that could do Intermodal um sensory things and memory and it was doing counterfactual simulations and it had motivated looking action and modulation of attention

Global availability of some variables or some processes and it was resilient within some resilience of course we're not resilient to getting hit with a hammer on the head we are resilient to you know a train coming at us from a mile away but not an inch away I mean it's always going to be about where you draw the bounds oh I mean that's out of bounds you don't need to be resilient to that kind of an insult to this drug but to this molecule you are so I would just ask if we had a robot that had some of these features would we include it in the conscious club or would denit include it in the club no would a pan psychist yeah they might say well it's something it's like to be that robot but where are we really getting if we just say it's something it's like to be this robot and this thermometer and this networking system well if the only way we can ask it is the exact same affordances that we as observers have today like measuring electromagnetic frequencies or something like that then aren't we going to to be debating these things forever because no one really disagrees that these things all appear to happen for systems that we think are complex gold directed systems like humans and cats and things like that but isn't the question about what else we're going to do Beyond this so hopefully that's where the projective geometry it's kind of like that's the uh hook that they hang their coat on so to speak they're going to bring it all to projective geometry and they're going to say this is what explains all these functional features and the variance which we'll get to in a second and it makes some other predictions that would be unique and testable so they're going to say yeah there's probably a lot of ways that theories of Consciousness could explain these features like attention schema Theory or the global workspace Theory or integrated information Theory all these different Frameworks for Consciousness but we're going to take these things and we're going to go in a New Direction by talking about the projective geometry so what supports in their eyes the need for not just a quantitative or a formal framework but also specifically the projective geometry and for them it's about these phenomenological invariants so it's kind of like Universal basic income it's like it's Universal you know these are invariant what part of that don't you get so these are the things that generalized systems of Consciousness you could debate with them whether these systems that they're talking about are the right ones or whatever but these are the invariant features of experience and phenomenology and they have them as relational phenomenal intentionality which is that all experience and all Consciousness is about relations especially between one and another system like internal and external relationship even though as we're going to kind of see it's a little bit trippier than that but basically it's about the relation of like the eye to not I it's like I thou I you these kinds of things there's a situated 3D spatiality and that means that you're starting you're in a situated

spot you're not just everywhere experiencing a 3D simulation but it's always perceived as from a situated spot in spatial dimensions three of them and then Through Time three and four there's this multimodal synchronic integration which is in a snapshot it's multimodal and then four is through time everything is happening so this is kind of capturing through space or through senses and then through time saying yep you're at a concert and there's the light and the sound and things bumping into you all of that is happening in this integrated way you don't just experience those separately um like at the very least least the illusion appears to cover integration across senses which are qualitatively different like a chemical is different than a photon from the Sun and then there's this temporal integration through time which is kind of related to the perceived event durations uh there's messages that can be flashed on a screen or sounds that can be outside of your uh awareness but there's also ones that can be within your awareness at a certain duration of time so if it flashes for a second it's able to be at a time scale that's integrated into Consciousness um there might be time scales that influence you below that but it doesn't become aware so like that's kind of conscious and subconscious but it's dealing with that um a little bit uh blurry about whether Consciousness is just the whole neuros system and anything that gets integrated into the nervous system through time is considered to be temporally integrated or is it that there's experiences that actually play out over certain time scales that we experience and then lastly kind of related to one is this subjective character which is that it's about the subject and one and five are very similar but in fact they hold that two through five are all implicated in one okay so they have the functional features that's pretty abstract and cybernetic and then they're like okay what are the invariants almost the axioms that we want to um be building from that we want our Theory to be able to address and then they're they lay them all out one through five and then they're like yep turns out that two through five all collapse back onto one so really one is the part we need to explain in how they've set up the problem so it's relational it's about the relationship between the agent and the world it's phenomenal it's based upon experience and it deals with intentionality and that is going to bring in agency affordances all these kinds of things that have come into play okay any thoughts on the invariance or the because in a sense that's what that's what frames the paper and it's what frames the discussion the kinds of answers they're going to come to is these are their checklist for how they'll know they'll succeed or this is the recipe these are the ingredients this is like what they're cooking with for the paper so if anyone thinks of it they can raise their hand I just want to clarify that because it's like you could think about this stuff all day but at least for a Journal Club it's like what did the authors actually specify to do did

they answer something that um yeah so Stephen um yeah I suppose also coming back to what Lee was saying about um perceiving things in terms of our cognition in relation to our history I when we started getting into perspective painting and thinking about things that way you get this sense of how much is it about us [Music] constructing cognition as a cultural way of being how much are we actually constructing this relational phenomenal intentionality culturally and that then because in indigenous cultures you see they think a lot more about shape shifting and the land being much more fluid and infolded um whereas we tend to have it much more linear and structured and straight edged so there's there's some question there about how much this Consciousness becomes like a construct that we actually put on the world yeah one um researcher who's done a lot there is the professor Tanya Lurman and collaborators at Stanford and she studies um like hearing voices in various uh cultures and so what is the difference between a society that says hey the voices that you hear are your ancestors and they're telling you cool stuff or it's a mole thing and it's all random or it's bad voices that are actually getting into your head that is going to be shading the experience of the people experiencing voices and so when we're talking about Consciousness what if one Society they tell everyone and multi-generations it's completely baked into their culture it's like yeah it's an illusion you know denit toia everybody it's an illusion and just the ideas that come to your head they're they're literally defined by being illusory and you're you're basically just in this Perpetual Auto gaslighting scenario just don't don't worry about the policies we're pushing on you just go with it and then another culture says actually the spark of awareness is what should be followed even if it appears to disassociate from social norms that's going to be a different outcome and so all of those different cultures could contain individuals that are 100% convinced that they're right so being right isn't the same thing or thinking you're right isn't the same thing as having an answer to the Consciousness question and yeah there's a lot of implications with how we actually are constructing our world and our perspective so that is very related to Lee's point I think he's Frozen for a second but yeah I agree that was a really good point about how um it's like we're in feedback we're doing Niche modification for how we perceive and what affordances we have in the Consciousness space if you train up people in a Insight Meditation tradition since they're children or something they're going to have affordances in their Consciousness uh it's another framework good or bad layer on top of that with the Judgment but they're depending on what people are taught about their consciousness is growing up that is going to color their experience and what affordances and what trajectories they take later on um so very interesting stuff and I wonder like what the authors think about that or how they

conceptualize these issues so just to cover this projective geometry bit because they WR the concept of 4D four-dimensional projective transformation is Central to the model as it yields an account of the link between perception imagination and multi-point of view action planning so they're going to try to link some of the cool stuff like we've talked about in this action perception Loop we've asked where is Imagination where is um Imagination and speculation and all these different things that we appear or experience engaging in and then also how do we Square the circle with the fact that we're so situated in our affordances yet we can also Imagine things that aren't seemingly uh related to our affordances like a city's grid it doesn't seem like it's tied directly to my affordances of course in a traveling sense it is but not like it doesn't depend on where I'm sitting in my building in a different city so geometry is about um the shape and the size of things whereas topology is about how they're connected so in topology we talk a lot about nodes and edges and about connectedness and structure whereas in Geometry structures related to angles and sides and edges and things like that and ukian geometry is kind of like linear algebra is a specific uh case of geometry where certain prerequisites are true like parallel lines don't intersect and the angles of a triangle are 180° there are non-ukian geometries just like there are non-linear models and non-ukian geometries which is kind of like the outgroup to the exclusion of ukian so it includes a vast number of esoteric geometries and the way that you can move from some geometries to other geometries is through a projection and that projection is called a map so it's kind of like seeing the map that's in front of you it's like it's a map it Maps your representation onto the territory so this is like a map that Maps one territory onto a different territory or one geometric projection of a territory onto another so you could map between two different kinds of maps of the world you can map between a flat one and a spherical one with certain distortions taken along and things like that but the key mathematical features are that in the ukian Paradigm there aren't special points um there aren't points at Infinity it just is like a vast plane that doesn't end there's no Horizon there might be a angle of vision that you can't resolve below but there isn't a horizon in a flat plane and then the origin is arbitrary in that every point you choose you could just say oh well this is you know 0 00 0 and the um local neighborhood is exactly the same here as it is everywhere else it's all square contrast it with non-ukian geometries which are like you can imagine though there are many non-an types have features that include things like special point point or points at Infinity like there is a ring of points at infinity and they're all equally infinite far away there can be Horizons there's um non-arbitrary Origins all this kind of stuff and they're basically going right the perspectival

phenomenal space so our perspective of what we're experiencing in a geometric understanding allows us to locate ourselves and navigate in an ambient space that is nevertheless essentially ukian so you can be in a 90° room but you can be having this like wacky distorted hyperbolic experience of this 90° room and kind of seamlessly integrating across that even if it was a bunch of Cubes that were connected you can climb through it you can navigate in this idian world and um measure the angles they're 90° it's all ukian but your perception is still going to have this distorted effect where things closer to you are um seemingly different than things further from you and that means that the angles as you look at them unless you're in a very specific Position will not look like 90° look around the room you're in even the corners don't look like 90° um and they basically say because um the ukian space has no privileged points no points at Infinity no Horizon no non-arbitrary origin it very strongly suggests that a non-ukian frame must be operating with a help of sensory motor calibrations to preserve the sense of invariance of objects across the multiple perspectives adopted and relate the situated organism to the ambient ukan space so like put your hand on a 90° angle you can pretty much feel from propri reception it's 90° you look down it's not 90° feels normal everything's chill because we're experiencing this Distortion of the ukian space okay any thoughts on the projective geometry or about geometry or where would geometry fit into a Consciousness model I just concur with them that there's some functional attributes that would Point towards non-ukian spaces but then every time like wait what in the how with geometry just how is it giving us experience again I guess that's sort of the uh the question but uh yeah Sasha um yeah I also thought this was really trippy um I really liked uh the 9.0 kind of pre uh pre-digestion of this uh topic so I found that really helpful um yeah I think this is kind of trying to walk us through um the possible you know points of view on uh Consciousness that um not only does each individual have a different perspective um necessarily in ukian space um like we're all seeing it from different locations let's say but that even the the mapping is different between individuals onto our own non-ukian you know projection of uh experience and so I I think that's that's kind of many steps to walk through to recognize how different um we see the world and how we can define this word perspective so I think that is really uh an important aspect of understanding Consciousness is uh where individual perspective is baked into the whole process yep and it makes me think about like um let me see it through your perspective or do you see it how I see it so we have visual metaphors for alignment and then we have put you know walk a mile in someone else's shoes or put yourself in their shoes those are embodied and active metaphors of understanding what else someone's experience is like and

then what we're saying here is um somehow we should should integrate that how you see it and how you walk it together so that your perspective is based upon both how you see it and the shoes that you're walking in so your actual embedded extended body and how you walk in those shoes and there's people for whom they can't walk a mile in any shoes so what are other metaphors that help us go beyond just simple walking to really understand that more generally we're talking about like perception and action and that because perception and action are linked in US in fact perspectively Linked In us it's also that way in someone else's so it's not just about coming to agreement with what they think or believe or state they believe nor is it about specifically reenacting their action sequences which we wouldn't either be able to do or wouldn't even have the same perspective on them as they do how are we going to collaborate and so there's like a lot of interesting stuff and I would hope that this provides a little bit more structure to sometimes discussions about everything from issues people disagree on to accessibility how could we whether we um bring in projective geometry or not how could we just start with the idea that everyone is going to have a unique perspective and then work for uh something Collective rather than get to this point where it's like well we disagree okay well now what um yeah Alex go ahead uh yeah I just wanted to say this this brings up uh it reminds me of general relativity in a way um like conscious is some sort of uh fundamental it's a if two folks are in different perspectives uh and they can observe the order of events occurring in different ways are in if they can uh observe events uh occurring out of order based on their perspective and a third remote uh thing that they're looking at uh maybe Consciousness is the way of uh self observant uh of a larger system when uh the system is essentially a manifold uh the Consciousness is the uh recognition of Relativity in actual space right um and this projective geometry seems to kind of be getting there um if we are unable to observe the entire world and we're only able to observe from our perspective uh Consciousness is the recognition that that is an a variant across space and other agents uh should have the same uh I don't know phenomena I apologize for terminality obviously no that's that's awesome I I really like this you you said the recognition of general relativity is consciousness and so in ukian bace it's like the invariant is angle's going to sum to 360 and this is going to work this way and the parallel okay so those are geometrical invariants but because the world is so quintessentially perspectival and embedded in an active actually the invariant is this relationship of um action and perception and that you move in a apparently non- ukian space because the invariant is actually like the locus of what is generating awareness is um Consciousness and then the two examples that came to mind are first off you

hear a noise with each ear at a different um time and you hear you know in photons probably know hit lights at your eyes at different times but also they see different things but you get a monocular perception from two different eyes and you know but you can push one eye you can violate an expectation and you can see two images but basically when it works it's monocular and then I was thinking of a concert where like the base is on one side and the treble speakers are on the other and then it's like the group they're actually hearing the events in a different order one person hears one note before the other one and vice versa but especially locally that doesn't mean it's incoherent so what is happening there and that's kind of like this group Consciousness question and just thanks for bringing it to the general relativity level because also with the general relativity um Innovation it was a movement from a cartesian Newtonian ukian all these things um movement towards like yeah maybe gravity isn't like a nonlinear Force operating on a ukian field but rather it is a non-ukian distortion of another thing and so there's really a lot to say there and then also there's like a lot of inroads to Quantum and Consciousness whether it's ham uh hammer off and Penrose and about the work on microtubules or whether it's just other more broader physics grounded theories there's so much there about where Quantum itself might fit in so a lot of points of contact there with non-ukian geometry perspective and um and then the fact that we don't overlap with any other system in terms of our perspective and so then how do we take that and it's like well just like you can build a computer in a world with general relativity can't we design information and sensemaking and attention systems that work in a world where people have different perspectives I'd hope so Stephen yeah I like this like really broadening the idea of relationships because I think what's really helpful with this diagram is it and talking about the map as a way of projecting between types of spaces rather than this idea of projection as representations in the mind being projected on the world which it's kind of the slight danger with that language you know because it gets used in that um in that rep representational way if it's somehow the way that we engage a type of spatial Arrangement almost bringing in spatial cognition work but bringing it much more into the body um and I think this relates maybe some of the work with Harry Heft he does some really interesting work in the world of spatial cognition um and it'd be interesting to see his perspective on this yeah what you said about projection so that's like in the Psychotherapy it's like you're not seeing it how it is you're projecting you're you're know you're you're um you're envisioning you're biased you're projecting onto the world you're projecting that emotions onto them but it's like what if you start from hey it is a projection you are in a projection you are the pro projection your

experience is the projection of all these things that we're talking about so it's like let's start from that point of view that you're not treating this person like a robot you're assuming they're having emotions so you're engaged in this relational projection that's intentional about your relationship so how could we connect for example optimism about the relationship to how you are going to map their facial expression onto your behavior not onto what do they think but on how you're going to act in a certain situation that's a pretty uh different context so I think that's like a very uh interesting to bring the psychological projection the almost the psychotherapeutic use of the term to geometric progression uh and and I'm sorry geometric projection and then connecting non-ukian geometric projections to to um general relativity and to control theory and about systems design all these different things so just a few quotes that we can kind of go through here um basically this is just a summary of what they believe the pcm's strengths are and how they summarize it which is that the structure geometrically of multimodal lived conscious experience closely approximates a projective three space so that's like a threedimensional space with kind of this geometer's lingo on which a projective transformation I believe pgl4 is saying it's a four-dimensional projection projected down but we could hear from a geometer with a little bit more information here or what would the physics Community notate this as our model fits intuitive data but it also helps account for other phenomena that might seem at first unconnected including working perspectival imagination inferences about the points of view and aims of others uh like the intentional stance modulo uh active inference and many peculiar phenomena associated with Consciousness so OB Out of Body Experience hiatos scopy D realization Oceanic and mystical experiences it also sheds light on some visual phenomena which we'll definitely get to and it sounds like there's a bunch of papes and they're probably writing more so if any of the authors want to you know share what they're thinking or doing now that's cool here is an example we talked about a little earlier which is that when we consider like think about the Empire State Building we think about it um in perspective like looking up at the P at the Empire State Building or just floating away far away so that it basically looks ukian because it's far away but it's going to be both conceptualized independently of any particular point of view like you could zoom in in your little brain AutoCAD simulator and imagine yourself on the 40th floor looking at a 90° angle in an ie beam and you wouldn't have any um deception that it's actually ukian 90° but then at the same time we can naturally co-experience almost this like looking up at it and understanding that no it's still not going to look 90° so that's what optimally allows with this projection the perspectival accessing of ukian models of the world

in memory for appraisal and updating of such models through active sampling so we get to benefit from the sort of statistical uh uh reality realism let's say at least not reality of having a model that's actually rect linear for example but then also have this total fluid projection that allows us to act and perspectiv perspectively understand ourself in that space so there's the relative perception and then there's that perspective independent objective knowledge based which doesn't mean it's correct it just means that it's about an object it's of an object it's not truth objectivity it's objective like it's about this box on my desk and productive geometry helps functionally connect subjectivity so being about the subject that's relational and then epistemic objectivity again not truth just knowledge about objects allowing us to see Consciousness as a kind of mediator between the situated organism and the objective World it inhabits very cool stuff they also right so Active Vision thus becomes like a palpation of a 3D spatial user surface to feel the epistemic affordances that are Quantified by expected free energy this anticipatory palpation with temporal depth requires a generative model that encompasses the consequences of actions and their effective values so in some other work by friston and colleagues where they've looked at for example reading and informational foraging with the ocular motor with the eye muscles and so the way that the eye scans around the page it's not based upon maximal reward it's based upon the maximally informative observation that could be made and what is that observation maximally informative in the framework of well top down priors about semantics so when somebody's coming towards you there's like subconscious icade movements that first assess for example the um broad category of person that it belongs to and then they will zoom in on different components of the face and culturally informed areas of the face that might signal different types of things like um whether there's going to be a good relationship in the uh future or not and so it is so interesting what the authors did which is to tie this information foraging optimal foraging info foraging info thermodynamics perspective take it from the cybernetics where it already had been very well entrenched and now say like that's our experience of vision that's why even when your eyes are cating at the level that you're actually perceiving it's actually a lot smoother because our experience is sort of smoothed out over those underlying Cades that being said those Cades are driven in a top- down way as a compromise with top down forces let's say by culturally informed priors your eyes won't know how to scan if you don't know how to read the language you won't know which parts the letter informative which parts the word but when you're fluent in the language then we know that like all the letters in the middle can be mixed up because you're really just doing a generative model of the text and that turns out that it includes

information about font and includes sentence structure all these other things you can't understand the sentence with the words mixed up in a different order if the letters are in the right order but the words are mixed the semantics don't allow the top down to inform the bottom up so your visual foraging is bunk because you're going to be looking at the difference between a b and a d but you're not going to understand what the sentence's meaning is because that's level we're actually at this framing of expected free energy through projective transformation in imagination provides a method for envisioning how things would look feel or be from another hopefully better location and thus guide action so like where would I be looking at if I were figuring out what this word were under my generative assumptions about language it is important to emphasize that according to the projective Consciousness model free energy minimization is always operating as the fundamental algorithm governing the field of Consciousness and thus there's never an end to the process as there is never in practice a solution that reduces free energy to zero which because it's defined as this informational distance against a non-stationary uh realism world uh it means that basically it's always going to be bounded by zero and he this I thought was very funny heel might be happy to know that on the projective Consciousness model Consciousness is thus always somehow unhappy even when it reaches the best possible States available to it so this is the uh wellknown hegelian dialectic of thesis antithesis synthesis often shown as a loop as if it were just this runaround rock paper scissor here it's actually like a virtuous or a vicious spiral and it just showing how this semantic ordering of events with thesis antithesis and synthesis it's like it never ends free energy minimization optimization cybernetic control processes multiscale evolutionary adaptive systems they don't end they can't take a break there can be relaxing times but they can't rest in the actual like stasis there can be stasis with respect to a certain Dimension but that's reflected by this seething activity that actually structures the strange attractor so that the system returns to these homeostatic preferred expected and optimal States and if you prefer over evolutionary time something that isn't optimal or isn't functional you're not going to exist that model will not exist of the statistical regularities in their Niche because their preferences the observations they seek out will include things like toxins or jumping off of cliffs these are not adaptive observations uh inter receptively or from a visual perspective and so you end up not seeing models that prefer those things and so it's like this endless preference hunt policy adaptive selection stuff happening um pretty cool stuff by the authors any thoughts before we just look through the figures before we close out see you sash thank you so just to spend a couple minutes on each figure and just see what they kind of do and then maybe we could

either reach out to the author or just everybody who's listening to this treat it as an open call to come participate if you're curious about this or something does or doesn't add up let's hear about it cool so figure one is this account of psychological phenomena and I think uh it's almost like Minecraft helps us understand their simulations so let's just imagine that they're doing this in Minecraft so in the a is like the Builder mode of Minecraft which ironically is still from a perspective even though they're trying their hardest to represent a ukan world it turns out that can only be done from like the position of a camera wherever that camera is there's no location independent representation of this ukian space unless they did like just a matrix and they said look just render it yourself but when you rendered it it would have to be from a perspective you can't have all the perspectives at once or flattening in a dimension that doesn't uh work like that but then the lived space S^2 which is um they're using geometric notation I'm not totally sure about it maybe again a geometer could come on but like basically what they're saying is even if you have this block of interest when you put a person's head there or their gener the locus of their generative model um they're going to see things in this distorted way B is the anti-space beyond the plane at Infinity so I don't know exactly what it is but it's and it was interesting they said with a transparent plane at Infinity so it's actually bringing this like transparent uh notion back but this is like it's the opposite of what you see in some way I I guess it's like it's the yeah it's everything what you don't see it's like when you're watching the sunset what is not being seen and maybe is the anti-space and then this um God's eye projection are like somehow a transformation of the space that includes a lot of the planes of symmetry they write almost all the space is visible but completely warped and the world model manifest as structures with complex symmetries so that's quite interesting because whether you agree with it being God's eye or not I would say it's just another projection of the system um or just another at least as we're seeing it but maybe it's in like a ton of Dimensions or it has like other trippy features but this is kind of like highlighting just the uh the the invariances of the visual field but that doesn't guide action like if you're a baseball player and you're running towards the ball maybe there's like you know the one symmetry would be all the grass when you look down and all the Blue Sky when you look up and then there's a ball and that's like breaking the Symmetry in some very nuanced way in a dynamic way but what do you want to actually see as the ball comes towards you well it's actually what we do see which is that the ball gets bigger and apparently more manipulatable and we can do increased resolution on things like it speed when it's further away that helps us um estimate where to go but it's close we can fine tune so it's like exactly what you would want as an embedded agent

is what we're experiencing and so we're not experiencing this like whoa it's just like hidden symmetries nor are we experiencing like the Minecraft or the Grand Theft Auto perspective we're in our own embedded situated experience relationally engaging with affordances and the field of Consciousness and the field of affordances are very related interesting stuff great representations at very least just stuff to think about and learn this one I thought was very informative so this is a que um we can think of this visual illusion as a q that's ambiguous so it's called The necker Cube and basically depending on how you see it cough cough you can either see the bigger wider side as like pointing up into the left which is left and right already are relational spatial and down and to the right so it's like you know which way let's just let's just have a quick show of hands raise your hand if in looking at either of both of these options one is going to be I see it um basically going up to the left two is down to the right and three is it switches for me okay option one who sees it go up and to the left primarily okay option two who sees it go down into the right primarily and then who sees it just like rapidly fluctuating okay yeah I'd say it seems like people raise their hand in every category small sample size but yes it's an illusion it's something that is ambiguous and the reason why is it's a three-dimensional Cube at least it suggests it hints at being a three-dimensional cube but of course it just lines on a page okay we unpack that in a very generous optimistic way as being a two-dimensional drawing of a cube if a child drew that you'd say the child is trying to draw a cube they're not drawing like a logo with like you know a double helix and two you know it suggests a cube it hints at a cube but it's a two-dimensional projection onto a piece of paper well duh but it turns out that without shading one of the sides or adding a light which implies a perspective it's like you can flip between these two solutions so to speak of how to represent it and so u in relation to this cube is actually this ambiguity that's being um flipped back and forth between and then specific um situations might cause it to um prefer one side or the other yeah Lee what do you think about that well I think this I mean I'm really interested in this because it kind of relates to um to the the point about unconscious simulations so if you're looking at it and and it flips which it can which it can do and it does do if you start to pay more attention to that you you realize that it doesn't just flip there's a sense in which there's um there's a kind of an unconscious doing which is probably I'm assuming to do with the predictions that you're making so you know if I pay really close attention what I notice is that I'm isolating uh or tracing maybe different portions of the cube in different ways and then labeling them in different ways so I kind of go oh that's a you know if this bit is a corner or that's the front or that's the back so and it's that that actually makes it kind of shift and I think

it points to this idea you know like of that the you know the perceptual world is is kind of a co-construction in a way but it's it's the it's that I suppose what I'm interested in is that it's a it's a kind of a doing you know there's a there's a way in which we're active in it which is more often than not transparent you know we don't necessarily so for example a more real world example might be um you know if we look at a tree we might label it an object you know typically we do but um but actually it's kind of more of a it's more of a process but we don't we don't see it that way because we just kind of you know we look at it and you know for all intents and purposes if it's not Salient to us we wouldn't notice that but anyway yeah well no it's super interesting and the semantics and the illusion it's like there's the old woman young woman illusion um and then there's like the Bunny and the uh duck and those the semantic level can be like I'm looking at a bunny and then you'll basically not see the duck as much okay so the semantics can inform us especially when you label for Action like this is the side with a handle all of a sudden be like no I can feel it dragging I'm only seeing now one of those two sides fixed I'm keeping my eyes right where I want the handle to be and all of a sudden it feels easier to keep it in one zone and then also it's like there's illusions that are super ambiguous for some people and other people like I'm sorry I just don't see the young woman in that illusion I just don't see the bunny I don't know what you're I don't see that dress as that color remember that whole thing so that's like because of the subconscious you know the iceberg of the actual neurophenomenology the underpinnings and the mechanisms some people are near a critical transition point in their perception where just by scanning their eyes the ballerina will flip one way or the other other people because of how their deep priors guide their visual foraging and their info gener and their information foraging and their by generation of model they just will never be able to have the ballerina switch directions never never be able to have the duck go one way if you haven't seen a duck you're not going to see it in the back of the head of a bunny yeah Stephen yeah I saw an interesting presentation by bur Lotto he does some work down in Neuroscience with cir to Cay actually and he showed this picture I think it was fairly similar picture and he said that we mostly see it from above because we expect to see things from above first which I thought was kind of interesting so there's kind of like you do end up with this kind of inactive situated piece because it's like well I'm sucking sunic here I'm probably looking from above if I'm looking from above these things pop out then well maybe it's um so you know I thought that was kind of that reminds me of children's cereal being put lower on the aisle to get them in their zone of affordances it's in their proximal zon so you can do niche-based partitioning of surveillance capitalism great I

love it and then also uh there's a lot of other uh natural situations where we can track what an animal looks at and yeah they're looking at things that are actionable and that's like us grabbing something from the top there's not maybe foraging for branches you don't visualize it from the top but that's something that we can do but really we specialize on picking up things that were above okay let's look at just this last two figures so figure three I think is uh related to this uh paper 2018 paper that they mentioned in the very end for more details on this family of simulations but essentially it's showing like multiple views of the uh expectation map and the prior belief map and basically an agent has to move from the left side to the right side so like it moves from here's its left here it starting and it has to move through its pit I don't know if that's a bad state or what about it but basically here it is blue it believes there's a goal over here it's going to pierce into this darker blue with its zone of prox awareness essentially move through the pit resolve uncertainty about what actions to take here the action path is simple just keep walking in a straight line um even though it looks like narrow and you're then finally it gets in contact with the uh goal State purple and so from its local geometrically uh distorted point of view projected point of view there's a free energy thermometer and it's like first it scans around and then it's like okay I'm going to go that way towards the goal and reduce my uncertainty and then it's just getting lit up and that's interesting in ly where dopamine comes into play it's something I would love to talk about soon because I've worked on dopamine the free energy principle perspective dopes usually thought about as a reward molecule oh it makes you happy or it's the reward you know slot machines and all that but if we think about it in this context dopamine firing patterns in this phase anticipatory of the reward actually have to do with free energy and um when you get to the reward your Precision is confirmed because you already believe there was a reward there so actually and this is shown in figure four as you approach the max zone of maximal information uncertainty reduction that's when the free energy is uh getting out of control so to speak and then when the goal is achieved Precision is achieved again so when we think about what the neurophysiology actually is and how different interventions in our neurophysiology might map on to differences in what about somebody who doesn't have motivation even though they know the goal is right here but they're just just not even if they can see there and other people say it's well lit you can just do it and they just don't have the motivation maybe someone else is here and they go actually I like the other side more or all these different things without passing judgment on what these states are what level of analysis or what culture says they're rewarded or not just how can we map some of these cybernetic parameters onto the neuro and eventually the phenomenological

parameters and I think figure four gets out a lot of similar ideas because it's talking about this relationship between the lived space and the peripersonal space one might almost be able to say and also events that are perceived or imagined and that surprisal and then reduction of uncertainty the sort of realignment reordering of either action or the inference side to reduce surprise about the world given this very deep generative model um that's kind of where it's at and those were some really cool ideas brought up in the paper and just if anyone else wants to raise their hand just a few more interesting notes on this figure I thought were cool they show a loop from um prediction to the uh perception to prediction of the change in free energy imagination and then action and so we've been talking a lot about cycles that go like perception action world right perception agent action world like that but here this is a loop that happens in an organism or in a model of analysis still it's a nested marov situation and it doesn't replace the other one it's not the other one the other one is about agents in the world and this is actually like saying in these reflexive conscious systems what is actually the loop that allows imagined events to influence action so I I don't know if this is and it shows which directions different things can influence each other here it has perception and action influencing each other so maybe that's the other world in the agent Loop but then here's these other processes playing out just very cool and then here's like the vector of time on a free energy landscape through time and it's projected and it's going downhill so to speak through time in this like vector weird way and as I understand it and I barely understand it that's where the math and the research is it's like okay chemical reactions always go downhill well a candle burning is still downhill okay what about a cell well a cell is also downhill in a really specific way well all of our policy actions are downhill in what landscape that's the free energy optimization how is the landscape changing that's a relationship of how are meta optimization proceeds amidst different kinds of environmental regularities so um yeah that's all I would say about these figures before we close anyone else have any last thoughts what what would be something to go into more in 9.2 yep let's do um Alex and then Stephen uh I'll try to be brief um uh it it's interesting to me like relating this to the free energy principle that you model these Dynamic systems as agents which are trying to uh export energy from their identity um uh by performing action in Greater system that they're a part of uh their action actually creates new unknowns that they have to perceive and solve uh and maybe cognition is just the boundary between direct feedback and the larger environment they are part of like maybe it is the self-referential like structure that necessarily creates itself cool I don't know cool awesome thoughts Stephen yeah just thanks for unpacking these diagrams because I I had looked at this

paper a fair bit once in the past but some of this stuff was really hard to so this is this is really helpful um I I think this idea of the that that there is this imagination that's going on is really useful and I I think I don't know how many other papers really deal with this so I think that's quite quite interesting and one bit in the paper that we might want to think about that I did notice is this this it does say in the paper Sun about how this doesn't try and deal with all the subconscious Dynamics underneath but that that is about trying to find this kind of extra layer of conscious imagination or projection so it might be interesting just to explore that next time cool thanks for the suggestion and everyone thanks a lot for participating we will have follow-up forms for the participants it's in your calendar invite and uh via that mechanism for the live participants or just a Twitter Youtube comment for anyone else please provide us with feedback suggestions questions we'll be able to go into them for 9.2 good groundwork today and then yeah 9.2 we'll be able to talk a lot more hopefully here from a bunch of different perspectives um and just stay in communication with Team Comm it was a great discussion really thanks everyone for participating it's just like super fascinating stuff so thanks again for the conversation and we will talk next week everybody who's hearing it you're welcome to join next week or the following week but thanks again everybody and we'll uh see you later oh